

KILN

Operations & Quality

The execution, quality and uptime specialist. Explains what the plant is making, whether it passes quality controls and whether machines can keep the schedule running.

Made for: product, engineering, operations and business teams.
Reading time: about 12 minutes.
Truth standard: describes the repository as implemented on 1 August 2026.

KILN

What KILN is here to do

A role, not an all-powerful chatbot

In one sentence

Explains what the plant is making, whether it passes quality controls and whether machines can keep the schedule running.

3

product areas grouped here

7

registered callable tools

4

allowed capability prefixes

0

agents it may delegate to

Receives

Planned make requirement · Material readiness · Production and inspection records · Asset downtime and PM signals

Produces

Execution and readiness evidence · Quality gap analysis · Reviewable audit/CAPA drafts · Operations work item

Allowed capability families

production.*

quality.*

maintenance.*

agent.*

The key distinction

The product-area grouping below shows where KILN contributes across XELOR. The callable-tool table later shows the smaller, exact surface Agent OS can invoke today. Product ownership does not automatically create runtime authority.

The business areas under KILN

Business ownership is broader than today's Agent OS tools

Production

Orders, snapshotted component requirements, issue, completion and finished-goods receipt.

QMS & Audit

Sampling, inspections, dispositions, evidence, CAPA and audit readiness.

Maintenance

Assets, work orders, downtime, preventive maintenance and reliability measures.

How to read this grouping

KILN is the named specialist when a mission needs facts or analysis from these areas. Each module still owns its own rules, records and write paths. The agent coordinates through registered services; it does not become the database owner.

What remains true underneath

- Each module keeps its own business rules and permissions.
- Cross-module work passes through narrow ports or registered capabilities.
- Tenant isolation, audit and workflow apply regardless of which agent is speaking.
- A readable agent answer never replaces the source record.

How KILN participates in a mission

The mission engine controls order, parallel work and recovery



1 · Scope

ONYX turns the goal into a versioned graph run with a fixed tenant, user, limits and expected outputs.

2 · Authorise

The capability registry checks that KILN is allowed to call the named tool; RBAC repeats the user permission check.

3 · Execute safely

The registered domain service performs the operation. The agent never receives a general SQL doorway.

4 · Preserve evidence

Inputs, outputs, node events and checkpoints are stored so the mission can be explained or recovered.

Delegation

KILN is a specialist and does not delegate to other agents in the current registry.

What KILN can call today

Every row is explicitly registered and permission-checked

Capability key	Mode	Result	User permission	Approval?
production.orders.read	Read	Production orders	production.order.read	No
quality.inspections.read	Read	Inspection state	quality.inspection.read	No
quality.evidence.collect	Analyse	Evidence register	quality.inspection.read	No
quality.audit-plan.draft	Draft	Audit plan	quality.inspection.read	No
quality.capa-plan.draft	Draft	CAPA plan	quality.inspection.read	No
quality.audit-pack.draft	Draft	Audit pack	quality.inspection.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

What “dispatch” means

If listed, `agent.action.dispatch` appends a governed work item for a named specialist. It does not directly change a sales order, stock balance, production order, journal, payment or customer message.

Read / analyse / simulate

Returns evidence or a reproducible view. No business record is changed.

Draft

Creates reviewable proposed content. A draft is never labelled as an approved or completed business outcome.

Execute

Effectful at the Agent OS level and therefore requires an approved ancestor node.

Permission

The agent's allow-list and the signed-in user's RBAC must both allow the call.

Who KILN works with

Specialists exchange evidence through ONYX, not hidden side conversations

AXLE

Returns real execution and capacity facts to the plan.

SPAR

Uses the controlled inventory path for component issue and finished stock.

MICA

Provides production and quality evidence behind a customer commitment.

A complete controlled mission



Why this structure matters

Each agent stays accountable for its own facts. ONYX combines them, HEXA checks control evidence, and a person controls the decision boundary. This prevents one fluent answer from quietly becoming authority over the whole company.

Can this production order finish and pass inspection?

A realistic presentation path from question to controlled outcome

1

KILN reads the production order and inspection state, then connects downtime or preventive-maintenance risk from its broader module ownership.

2

Quality calculations and lot verdicts are code-based. KILN can collect evidence and create drafts, but a draft is clearly labelled and remains reviewable.

3

Production cannot mark a controlled completion as good stock unless the required quality gate and inventory posting rules are satisfied.

What the presenter should say

“KILN contributes verified domain evidence, with source references and limitations. It does not claim that a proposed or dispatched action is complete until the controlled system records and verifies the outcome.”

Useful questions to ask during the demo

- Which source records support KILN's answer?
- What permission was checked?
- Is this a fact, a simulation, a draft or a completed result?
- Would this step wait for human approval?
- What happens if a node times out or the service restarts?

How KILN stays trustworthy

Controls are enforced in the runtime and modules, not left to prompt wording

Control 1

Production uses a BOM snapshot for reproducibility

Control 2

Quality limits and sampling context are snapshotted

Control 3

Issued stock and dispositions require traceable stock references

Control 4

Asset downtime intervals cannot overlap

Identity boundary

Verified user + tenant

Authority boundary

Agent allow-list + user RBAC

Action boundary

Approval ancestry for side effects

Data boundary

Domain service + tenant-fenced database

Evidence boundary

Append-only run, event and audit records

What is live—and what is not

This page prevents the demo from overclaiming

Live in the repository

- Production-order and inspection reads
- Quality evidence and drafting tools
- Quality-gated production
- Downtime/PM foundation
- Operations contribution to missions

Deliberate current limits

- Maintenance reads are not yet exposed as separate Agent OS capabilities
- Draft plans and packs are not approvals
- No autonomous production, disposition or maintenance closeout

Runtime disclosure

Orchestration, ERP reads, approval gates and governed action dispatch are live. Language reasoning is deterministic; no external model API or connector is active.

Primary implementation references

`apps/api/src/modules/production/`

`apps/api/src/modules/quality/`

`apps/api/src/modules/maintenance/`

This guide is a plain-language operating map, not an API contract. The code, migrations and automated tests remain the binding technical source.

KILN in one page

Use this page when explaining the agent to a new team member

Job

Explains what the plant is making, whether it passes quality controls and whether machines can keep the schedule running.

Owns

Production, QMS & Audit, Maintenance

Receives

Planned make requirement · Material readiness · Production and inspection records · Asset downtime and PM signals

Returns

Execution and readiness evidence · Quality gap analysis · Reviewable audit/CAPA drafts · Operations work item

Can call

production.orders.read, quality.inspections.read, quality.evidence.collect, quality.audit-plan.draft, quality.capa-plan.draft, quality.audit-pack.draft, agent.action.dispatch

Cannot do

Maintenance reads are not yet exposed as separate Agent OS capabilities · Draft plans and packs are not approvals · No autonomous production, disposition or maintenance closeout

Plain-language glossary

Agent	A named specialist role with an allow-listed set of tools—not a free-roaming process.
Capability	One registered operation with an input, output, owner, mode and required permission.
Graph	A versioned mission recipe showing order, parallel steps, joins and approvals.
Checkpoint	A stored progress snapshot used for explanation and safe recovery.
Governed work item	An append-only assignment created after approval; it is not the final business mutation.